

Routing

Network routing is the process of selecting the best path for data to travel across one or more networks. It plays a vital role in the internet by ensuring that data packets reach their correct destination efficiently.

- Routing determines the path IP packets follow from source to destination. •

It is used in packet-switched networks like the Internet.

- Different routing methods are used to optimize speed, reliability, and performance.

Router

A router is a networking device that controls how data moves between different networks by checking destination IP addresses and choosing the best path.

- Forwards data packets between networks
- Works at Layer 3 (Network Layer) of the OSI model
- Selects the best route for data transfer

Different Types of Routing

Routing is typically of 3 types, each serving its purpose and offering different functionalities.

1. Static Routing

Static routing is a non-adaptive routing method where routes are configured manually by the network administrator. It provides complete control over routing decisions but is best suited for small networks.

- Routes are added manually
- Full control over packet routing
- Simple but hard to manage in large networks

2. Dynamic Routing

Dynamic routing is an automatic and adaptive routing method where routers choose paths using algorithms. It is widely used in modern networks due to its flexibility.

- Routes update automatically
- Adapts to network changes
- Suitable for large networks

Working Principle of Routing

Routing works by finding the shortest path from the source node to the destination node across a network.

Here's the step-by-step working of routing:

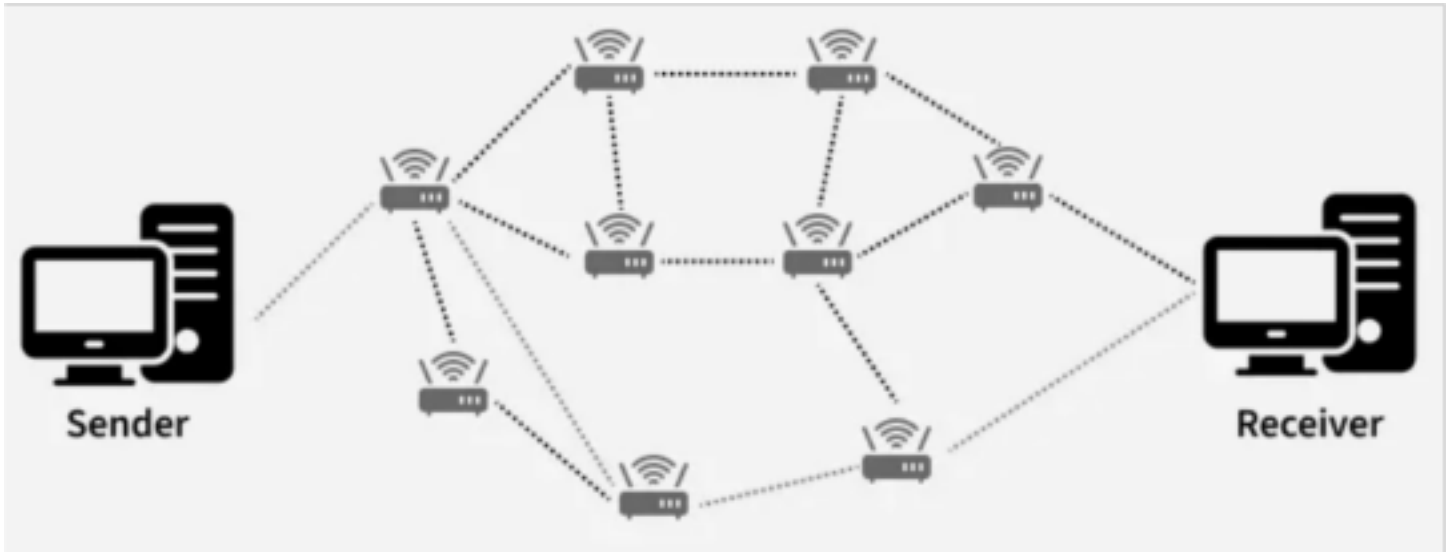
Step 1: Communication Start - A device (client or server) starts communication using the HTTP protocol.

Step 2: Data Packets: The data is broken into small packets. Each packet is tagged with the destination IP address.

Step 3: Routing Table - Routers use a routing table to find the best and shortest path to the destination.

Step 4: Hopping - Packets travel through multiple routers (hops) to reach the destination. There is a limit on the number of hops; if exceeded, the packet is dropped and retransmitted.

Step 5: Destination - Packets reach the destination, are reassembled, and checked for errors.



In the above image, we have 3 major components

- Sender
- Receiver
- Routers

The shortest path is highlighted in red, the path with the least hop count. As we can see, there are multiple paths from source to node but if all the appropriate metrics are satisfied, the data packets will be transmitted through the shortest path (highlighted in red).

Main Routing Protocols

- **RIP (Routing Information Protocol):** It is a distance-vector protocol that uses hop count as a metric.

- **OSPF (Open Shortest Path First):** OSPF is a link-state protocol that finds the shortest path using the Dijkstra algorithm.

Different Routing Metrics

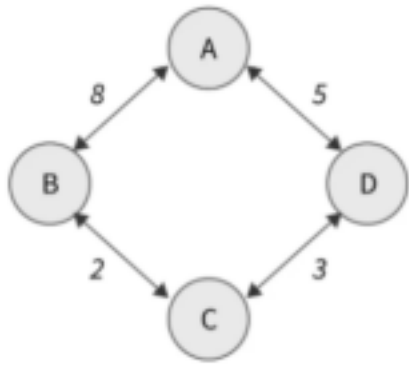
The purpose of routing protocols is to learn about all the available paths to route data packets, build routing tables, and make routing decisions based on specified metrics. There are two primary types of routing protocols, the rest of them ideate from these two only.

1. Distance Vector Routing

In this type of routing protocol, all the nodes that are a part of the network advertise their routing table to their adjacent nodes (nodes that are directly connected) at regular intervals. With each router getting updated at regular intervals, it may take time for all the nodes to have the same accurate network view.

- Uses a fixed length sub-net, not suitable for scaling.
- Algorithm used: Bellman Ford Algorithm to find the shortest path.

Example:



Step - 1

As we can see in the above diagram of the DVR network, the routers in the network start sharing their information with the neighboring routers in the network.

Routing table of A:

destination	distant	Hop
A	0	A
B	8	B
C	infinity	-
D	5	D

Routing table of B:

Destination	distant	Hop
A	8	A
B	0	B
C	2	C
D	infinity	D

Routing table of C:

Destination	distant	Hop
A	infinity	-
B	2	B
C	0	C
D	3	D

Routing table of D :

Destination	distant	Hop
A	5	A
B	infinity	B
C	3	C
D	0	D

Step - 2

After creating the separate local table this information is shared with the neighboring node having a direct link.

For Router A:

The router A has a direct connection to neighboring routers B and D.

Destination	Vector B	Vector D
A	8	5
B	0	infinity

C	2	3
D	infinity	0

So based on the vector information of the neighboring router, the value of the router is updated.

Here the optimal distance is calculated to reach the specific destination. First, the distance from A to B is identified, in our case which is 8, ie

$$\text{cost}(A \rightarrow B) = 8$$

$$\text{cost}(A \rightarrow B) = 8, \text{ and}$$

$$\text{cost}(A \rightarrow D) = 5$$

$$\text{cost}(A \rightarrow D) = 5$$

The distance to reach a destination B from router A is:

$$A = \min\{ (A \rightarrow B + B \rightarrow A), (A \rightarrow D + D \rightarrow B) \} = \min\{8+0, 5+\text{infinity}\} = \min\{8, \text{infinity}\} = 8 (B)$$

Since the cost is min from neighbor B so the router chooses the path from B. It then updates the routing information with entries

$$(8, B)$$

$$(8, B).$$

To find a cost to reach destination C from router A we will use a similar approach.

$$A = \min\{ (A \rightarrow B + B \rightarrow C), (A \rightarrow D + D \rightarrow C) \} = \min\{8+2, 5+13\} = \min\{10, 18\} = 10 (C)$$

Distance to reach destination D from A

$$A = \min\{ (A \rightarrow B + B \rightarrow D), (A \rightarrow D + D \rightarrow D) \} = \min\{8+\text{infinity}, 5+0\} = \min\{\text{infinity}, 5\} = 5 (D)$$

Consequently, A's new routing table is:

Destination	distant	Hop
A	0	A
B	8	B
C	8	D
D	5	D

For router B:

Router B receives information from A and C.

The new routing table for B is calculated as:

Destination	Vector A	Vector C
A	0	infinity
B	8	2
C	infinity	0
D	5	3

The cost for vector (B->A)=8

The cost for vector (B->C)=2

Distance to reach destination A from router B:

$B = \min\{(B \rightarrow A + A \rightarrow A), (B \rightarrow C + C \rightarrow A)\} =$

$\min\{8 + 0, 2 + \text{infinity}\} = \min\{8, \text{infinity}\} = 8(A)$

Distance to reach destination C from router B:

$B = \min\{(B \rightarrow A + A \rightarrow C), (B \rightarrow C + C \rightarrow C)\} =$

$\min\{8 + \text{infinity}, 2 + 0\} = \min\{\text{infinity}, 2\} = 2(C)$

Distance to reach destination D from router B:

$B = \min\{ (B \rightarrow A + A \rightarrow D), (B \rightarrow C + C \rightarrow D) \} =$

$\min\{ 8+5, 2+3 \} = \min\{13, 5\} = 5 (C)$

Consequently, B's new routing table is:

Destination	distant	Hop
A	8	A
B	0	B
C	2	C
D	5	C

For router C:

The router C receives information from B and D.

The new routing table for C is calculated as:

Destination	Vector B	Vector D
A	8	5
B	0	infinity

C	2	3
D	infinity	0

The cost (C→B) = 2

The cost(C->D) = 3

Distance to reach destination A from router C:

$B = \min\{(C \rightarrow B + B \rightarrow A), (C \rightarrow D + D \rightarrow A)\} =$
 $\min\{2+8, 3+5\} = \min\{10, 8\} = 8(D)$

Distance to reach destination B from router C:

$B = \min\{(C \rightarrow B + B \rightarrow B), (C \rightarrow D + D \rightarrow B)\} =$
 $\min\{2+0, 3+\text{infinity}\} = 2(B)$

Distance to reach destination B from router C:

$B = \min\{(C \rightarrow B + B \rightarrow D), (C \rightarrow D + D \rightarrow D)\} =$
 $\min\{2+\text{infinity}, 3+0\} = 3(D)$

Consequently, C's new routing table is:

Destination	distant	Hop
A	8	D
B	2	B
C	0	C
D	3	D

For router D:

The router D receives information from A and C.

The new routing table for D is calculated as:

Destination	Vector A	Vector C
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A	0	infinity
B	8	2
C	infinity	0
D	5	3

The cost(D->A) = 5

The cost(D->C) = 3

Distance to reach destination A from router D:

$D = \min\{(D \rightarrow A + A \rightarrow A), (D \rightarrow C + C \rightarrow A)\} = \min\{5 + 0, 3 + \text{infinity}\} = \min\{5 + \text{infinity}\} = 5(A)$

Distance to reach destination B from router D:

$D = \min\{(D \rightarrow A + A \rightarrow B), (D \rightarrow C + C \rightarrow B)\} = \min\{5 + 8, 3 + 2\} = \min\{13, 5\} = 5(C)$

Distance to reach destination C from router D:

$D = \min\{(D \rightarrow A + A \rightarrow B), (D \rightarrow C + C \rightarrow B)\} = \min\{5 + \text{infinity}, 3 + 0\} = 3(C)$

Consequently, D's new routing table is:

Destination	distant	Hop
A	5	A

B	5	C
C	3	C
D	0	D

2. Link State Routing

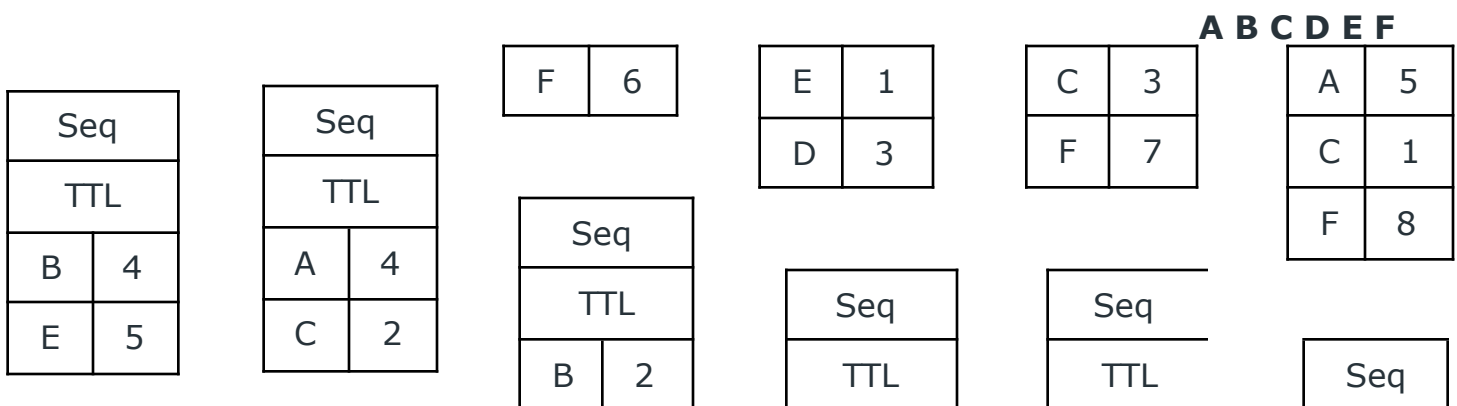
Link State Routing is a type of dynamic routing protocol where routers send updates only when there is a change in the network, saving bandwidth.

Routers share information about links such as cost and hop count to find the best path.

- It uses Variable Length Subnet Mask (VLSM) for better and scalable addressing.
- It uses Dijkstra's algorithm to calculate the shortest path.

Example:

Step 1 Prepare the link state packet at every router.



TTL

E	8
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B	6
---	---

D	7
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Step 2: Use Dijkstra Algorithm to find shortest path from A to other nodes.

A Table	B	C	D	E	F
AB	4	∞	∞	5	∞
ABE	4	6	∞	5	10
ABEC	4	6	∞	5	10
ABECD	4	6	9	5	10
ABECDF		6	9	5	10

A → B = 4

A → C = 6

A → D = 9

A → E = 5

A → F = 10

Advantages of Routing

- **Routing:** Helps send data efficiently by choosing the best path.
- **Automated Routing:** Uses algorithms to reduce manual work but gives less direct control.
- **Manual Routing:** Provides better control but requires more effort to manage.
- **Scalable Routing:** Works well for large networks and supports load balancing to reduce traffic and data loss.

Disadvantages of Routing

- **Static Routing:** Suitable only for small networks; difficult to manage and less secure in large networks.
- **Dynamic Routing:** Offers less manual control and uses more bandwidth and system resources.
- **Default Routing:** Easy to use but can cause issues if the default path is not properly configured.